CHAPTER 6

Physiology and Psychology

THE 19TH CENTURY WAS A TIME OF GREAT CHANGE IN EUROPE AND America. Agricultural reforms ensured a steady food supply, and improvements in public hygiene decreased fatalities due to contagious diseases such as cholera. The population of Europe increased from about 140 to 420 million people between 1750 and 1900, with many congregated in the new urban centers. The dramatic expansion of industry led to a general increase in wealth, although the insecurities of the capitalist state (with periods of boom followed by periods of economic downturn) led some to question a system in which most of the wealth was owned by a privileged few and to look to alternative political systems such as socialism and communism. New developments in transportation and communication saw the spread of modern road networks, railways, canals, ocean lines, and telegraph and postal systems (Jansz, 2004).

The 19th century witnessed the growth and increasing political strength of the middle class, whose long struggle to attain voting rights eventually bore fruit, although throughout most of the 19th century real political control remained in the hands of the conservative aristocracy. In the reactionary period following the Napoleonic wars in Europe, which ended with Napoleon's defeat at the battle of Waterloo in 1815, naturalistic approaches to psychology were repressed through censorship and the secret police. Nobody who promoted such views could hold a professorship in Europe and America in the early half of the century, and in the years immediately following 1815, advocacy of such views was punishable by imprisonment in some parts of Europe (Reed, 1997).

Joseph Priestley (1733–1804) was hounded out of Britain for his promotion of Hartley's associationist psychology as materialist psychology. Erasmus Darwin (1731–1802), the grandfather of Charles Darwin (1809–1882), who developed an early naturalistic evolutionary theory in *Zoonomia* (1794–1796), found his work suppressed. One of Darwin's followers, the British surgeon William Lawrence (1783–1867), published his theory that insanity is a neurophysiological disorder in *Lectures on Physiology, Zoology and the Natural History of Man* (1819). The medical establishment forced him to withdraw his book, and he lost his lectureship at the Royal College of Surgeons (Reed, 1997).

After the (failed) European revolutions of 1848, a new alliance of the conservative aristocracy and the middle class implemented a variety of reforms and ceased to depend upon traditional Christianity as the foundation of the social order. Although religion remained a conservative force in politics and education, the 19th century saw the emergence and general acceptance of more secularized systems of thought. While naturalistic treatments of physiology and psychology still stimulated vigorous reaction from the clergy, many came to see the development of science as independent of religion. Many 19th-century physiologists and psychologists avoided conflict with organized religion by maintaining that their theories had no implications for theology, since they held that questions about God are beyond the realm of scientific knowledge.

In the late 18th century, the work of Antoine Lavoisier (1743–1794) and John Dalton (1766–1844) had set chemistry upon a sound experimental footing. This stimulated 19th century physiologists to explore the physics and chemistry of organic structures and processes, including the structure and function of the nervous system. The 19th century witnessed major advances in neuroanatomy and physiology, which played a significant role in shaping the development of scientific psychology in Britain, Germany, and America.

POSITIVISM

Auguste Comte (1798–1857) introduced the term **positivism** to describe his view that the highest form of human knowledge is knowledge of the correlation of observables. In his **law of three stages**, he claimed that societies pass through three stages of cognitive development that represent different attitudes toward the explanation of natural events. In the **theological stage**, natural events are explained in terms of anthropomorphized forces; for example, lightning storms are explained in terms of the anger of the gods. In the **metaphysical stage**, natural events are explained in terms of depersonalized forces; for example, planetary motions are explained in terms of gravitational forces. In the **positive stage**, natural events are explained in terms of the description of observable correlation, which can be employed to predict the course of nature.

Comte believed that the natural sciences had developed systems of positive knowledge and that a similar approach should be applied to the science of society. A science of sociology (the term was coined by Comte) would ideally establish a system of laws describing regularities in human behavior. According to Comte, these laws could be employed to create a perfect society based upon scientific sociology, in contrast to the misery and chaos that were the natural outcome of social systems grounded in metaphysical speculation and religious superstition. He believed that a social and political system based upon scientific sociology would eventually displace traditional religion and politics.

Comte was greatly affected by the political upheavals following the French Revolution. He became secretary to the social theorist Henri Saint-Simon (1760–1825) in 1817 and supported himself in later years by private teaching and public lectures. He produced the six-volume *Cours de Philosophie Positive* during the years 1830 to 1842 and the four-volume *La Systeme de Politique Positive* during the years 1851 to 1854. His early work attracted many supporters, but he alienated many of them when he developed a new scientific religion based upon positivist principles. He set himself up as its pope, with his mistress substituting for the Virgin Mary (Reed, 1997).

The form of positivism that many 19th-century theorists embraced was the dogmatic empiricism advocated by Berkeley and Hume, shorn of their philosophical idealism and skepticism. According to this atheoretical form of empiricism, scientific theories and causal explanations are restricted to the description of the correlation of observables, which enable us to predict and control nature. In this view, we have no knowledge of real efficient causes or final causes, such as the nature of gravity or purposive design in nature. In *Auguste Comte and Positivism* (1866*)*, John Stuart Mill characterized the basic principles of Comte's positive philosophy in the following dogmatic empiricist fashion:

> The constant resemblances which link phaenomena together, and the constant sequences which unite them as antecedent and consequent, are termed their laws. The laws of phaenomena are all we know respecting them. Their essential nature, and their ultimate causes, either efficient or final, are unknown and inscrutable to us.
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> —(1866/1961, p. 6)

Although Comte endorsed dogmatic empiricism, his positivist system differed from British empiricism and associationist psychology in two respects. First, he treated the publicly observable properties of physical objects as the subject matter of scientific knowledge, rather than the privately introspectable sensations and ideas favored by British empiricists and associationist psychologists. Comte was contemptuous of introspection as a scientific method and denied the possibility of a psychology based upon it.

Second, Comte placed sociology at the pinnacle of his presumed hierarchy of scientific disciplines, with physics at the base and biology in between. However, he left no room for psychology as an autonomous science of consciousness or behavior, located between sociology and biology. For Comte, the whole content of psychological knowledge was exhausted by sociology, which studies behavior in its social context, and phrenology, the branch of biology devoted to the correlation of functionally characterized behavior (such as aggressive or amative behavior) with discrete psychological faculties located in specific regions of the brain.

Later positivists, such as Ernst Mach (1838–1916) in *The Analysis of Sensations* (1886) and Richard Avenarius (1843–1896) in *Critique of Pure Experience*

(1888–1890), followed earlier British empiricists in maintaining that the correlation of private sensory experience constitutes the observational subject matter of scientific knowledge. Thus, all positivists maintained that the correlation of observables is the subject matter of scientific knowledge, but differed as to whether publicly observable properties of physical objects or privately introspectable sensory experience constituted the observational subject matter of scientific knowledge. For this reason one can characterize both the experimental science of consciousness developed by Edward B. Titchener (1867–1927) at the end of the 19th century and the behaviorist psychology developed by John B. Watson in the early 20th century as systems of positivist science. The former was restricted to the description of the correlation of private mental states, and the latter to the description of the correlation of publicly observable stimuli and behavior.

Throughout the 19th century a great many scientific theorists avowed some form of positivism, including such figures as John Stuart Mill, William James (1842–1910), Thomas Huxley (1825–1895), Franz Brentano (1838–1917), Emil du Bois-Reymond (1818–1896), Sigmund Freud, and Edmund Husserl (1859–1938). Yet often this amounted to little more than a commitment to methodological empiricism, the view that scientific theories must be based upon observation. Many avowed positivists freely speculated about unobservable states and processes, including unconscious mental states and processes.

While this no doubt caused some confusion, it served a useful purpose in the development of 19th-century science, including physiology and psychology. For whatever they took to be the observational foundation of science, and however strict or loose their approach to theories about unobservable states and processes, most positivists and empiricists were committed to the principle that knowledge of real efficient and final causes (such as the nature of gravity, the human will, and the purpose of God's creation) is beyond the realm of science. This principle was occasionally employed to disparage religion, but more often than not it was advocated as a means of peaceful rapprochement with theologians, many of who came to agree that the realms of science and religion should be treated as distinct.

Thus, for example, when the evolutionary psychologist Herbert Spencer (1820–1903) was accused of promoting atheism in his *Principles of Psychology* (1855), he responded by withdrawing from the realm of religious debate:

> Not only have I nowhere expressed any such conclusion, but I affirm that no such conclusion is deducible from the general tenor of the book. I hold, in common, with most who have studied the matter to the bottom, that the existence of a Deity can neither be proved or disproved.
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> —(Cited in Reed, 1997, p. 159)

The positivist and dogmatic empiricist claim that causal knowledge amounts to nothing more than knowledge of observable correlation also provided physiologists and psychologists with a convenient parallelist defense against charges of materialism, by enabling them to maintain that they were merely studying the physiological correlates of mental states, without speculating about the relation between them. While detailing the neurophysiological substrate of mentality in the "cortical grey matter of the brain," the 19th-century British neurophysiologist John Hughlings Jackson (1835–1911) claimed

> We cannot understand how any conceivable arrangement of any sort of matter can give us mental states of any kind. . . . I do not concern myself with mental states at all, except indirectly in seeking their anatomical substrata. I do not trouble myself about the mode of connection between mind and matter. It is enough to assume a parallelism.
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> —(1931, 1, p. 52)

ASSOCIATIONIST PSYCHOLOGY

The tradition of associationist psychology initiated by Hume and Hartley in the early 18th century continued apace in the late 18th century and into the 19th. It was developed by a variety of British theorists, such as Abraham Tucker (1705–1774), who tried to derive principles of morality from associationist laws; Archibald Alison (1757–1839), who tried to account for aesthetic feelings in terms of association; Thomas Brown, who developed a number of "secondary" laws of association or "suggestion"; and George Henry Lewes (1817–1878), who extended association to accommodate logical reasoning, by developing "laws of thought" based upon a "logic of signs." In France the work of the sensationalists and ideologists was extended by M. F. P. G. Maine de Biran (1766–1824), Pierre Maurice Mervoyer (1805–c. 1866), and Hippolyte Adolphe Taine (1828–1893). James Mill (1773–1836), John Stuart Mill, Alexander Bain (1818–1903), and Herbert Spencer (whose contribution is considered in chapter 7) introduced the most significant modifications of associationist psychology.

James Mill: Points of Consciousness

James Mill, the Scottish philosopher and economist, was a close friend of Jeremy Bentham (1748–1832), the founder of **utilitarianism**. According to utilitarian theory, moral, social and political questions should be determined by the principle of utility: The right course of action in any situation is the one that maximizes human happiness and minimizes human misery. Mill was an early 19th-century radical who advocated utilitarian positions on government, jurisprudence, and education. Like Bentham, he supported a variety of interventionist social programs such as state education, health care, and poor relief, which he believed were justified in terms of their contribution to human happiness and the alleviation of human misery.

Mill published *A History of British India* in 1818, which provided him with entry to a successful career in the East India Company. His contribution to associationist psychology was his *Analysis of the Phenomena of the Human Mind* (1829). In this work he characterized the sensory elements out of which ideas and associations are supposed to be formed as atomistic **points of consciousness**. Mill added little to the basic principles of associationist psychology developed by Hume and Hartley and claimed that the main purpose of his work was to document further evidence for the principles of association. However, he did claim that the principle of similarity is not a fundamental law of association, since he believed that it could be derived from the principle of contiguity.

Mill's interest in associationist psychology was secondary to his political and educational projects, and his main concern was to adapt associationist psychology to utilitarian social goals. He maintained, for example, that a major task of education is to facilitate the association of individual happiness with benevolent social behavior.

John Stuart Mill: Mental Chemistry and Unconscious Inference

John Stuart Mill was lucky to survive his father's intensive private education, based upon associationist psychology and utilitarian principles. The young Mill was introduced to Greek at the age of 3, to Latin and mathematics at age 6, to philosophy at age 8 and logic at age 12. In his teenage years he studied economics and politics and prepared for a career as a lawyer, but eventually followed his father into the East India Company. A nervous breakdown at age 20 forced him to reevaluate his personal and political orientation.

Mill developed his own version of the utilitarian "greatest happiness" principle in *Utilitarianism* (1863). His moral and political views were tempered by his association with Harriet Taylor (1807–1858), who was married with two children (and another on the way) at the time they began their relationship. Mill scandalized many of his colleagues by practicing (with the approval of Taylor's husband) one of those "experiments in living" that he advocated in *On Liberty* (1859). After the death of Taylor's husband, the couple married. Taylor's influence inspired Mill's pioneering feminist tract *The Subjection of Women* (1869), which he dedicated to her, as well as his unsuccessful attempt to introduce legislation on female suffrage.

In 1843 Mill published *A System of Logic*, in which he described the methods of comparative causal analysis known as the methods of agreement, difference, and concomitant variation, now commonly characterized as Mill's methods. He claimed that these methods provide not only a means of generating hypotheses, a logic of discovery, but also a means of evaluating hypotheses, a logic of justification. Mill agreed with Whewell and Herschel that scientific hypotheses need not be generated inductively (on the basis of systematic observations), but may be the product of creative inspiration. However, he insisted that scientific hypotheses, however generated, could be verified only by observations made in accord with the methods of agreement, difference, and concomitant variation.

Mill was an early supporter of Comte's positivist philosophy and arranged for his *Cours de Philosophie Positive* to be translated into English. However, his own endorsement of positivism amounted to little more than an endorsement of methodological empiricism. He championed the view that science is ultimately grounded in the correlation of observables, but he did not feel obliged to restrict science to the mere description of observational correlation. He was quite prepared to advance hypotheses about unobservable states and processes, including unconscious mental states and processes.

Psychological Science In *A System of Logic*, Mill characterized psychological and social sciences, which he called "moral sciences," as a "blot on the face of science." He maintained that "the backward state of the moral sciences can only be remedied by applying to them the methods of physical science, duly extended and generalized" (1843/1973–1974, p. 833), by which he meant his own methods of agreement, difference, and concomitant variation.

For Mill, a scientific discipline of psychology based upon the methods of agreement, difference, and concomitant variation could establish a system of associationist laws:

> The subject, then, of psychology, is the uniformities of succession, the laws, whether ultimate or derivative, according to which one mental state succeeds another; is caused by, or at least, is caused to follow, another.
>
> —(1843/1973–1974, p. 852)

However, Mill was realistic about the predictive possibilities of such a science. Although he thought it possible to determine the fundamental laws of association, he believed that practical prediction in psychology is limited by the difficulties of anticipating all the factors involved in human thought and behavior (1843/1973–1974, p. 554). For this reason, Mill maintained that psychology is bound to remain an inexact science, at least outside of controlled experimental situations. He also denied the standard Newtonian assumption about the universality of causal explanation and maintained that many psychological phenomena have a plurality of causes. He avoided speculation about the neurophysiological basis of mental states and processes, referring such matters to his friend and colleague Alexander Bain (Bain, 1855, 1859). Mill also championed the scientific study of character, which he called **ethology**. He conceived of character as a set of social capacities and propensities, which he suggested could be derived from the fundamental laws of association. However, he also deferred this task to Bain, who made a brave attempt in *On the Study of Character, Including an Estimate of Phrenology* (1861), probably the least successful of Bain's works.

Mill's primary contributions to associationist psychology were his *System of Logic,* his 1865 *Examination of Sir William Hamilton's Philosophy,* and his edited edition of James Mill's *Analysis of the Phenomena of the Human Mind* (1869). Mill reiterated the basic principles of associationist psychology detailed by his father, although he reintroduced the principle of similarity as a fundamental rather than a derived law.

Mill also questioned the universality of the aggregative account of concept formation common to most forms of associationist psychology. He claimed that the properties of complex ideas or concepts are often more closely analogous to the emergent properties of chemical bonding than the additive properties of mechanical combination and represented some associative processes as a form of **mental chemistry**:

> The laws of the phenomena of mind are sometimes analogous to mechanical, but sometimes also to chemical laws. . . . Our idea of an orange really *consists* of the simple ideas of a certain color, a certain form, a certain taste and smell, &c., because we can by interrogating our consciousness, perceive all these elements in the idea. But we cannot perceive, in so apparently simple a feeling as our perception of the shape of an object by the eye, all that multitude of ideas derived from other senses, without which it is well ascertained that no such visual perception would ever have existence. . . . These therefore are cases of mental chemistry: in which it is proper to say that the simple ideas generate, rather than that they compose, the complex ones.
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> —(1843/1973–1974, pp. 853–854)

Unconscious Inference Although Mill's notion of mental chemistry was not developed for this purpose, he employed something very close to it in his response to the challenge posed by Samuel Bailey's (1791–1870) critique of Berkeley's theory of distance perception (Bailey, 1842, 1843). According to Berkeley's theory (Berkeley, 1709), our perceptual judgments about distance are based upon learned associations between visual and tactile sensations. Mill defended Berkeley against Bailey, but in the course of so doing was forced to revise the basic assumptions of associationist psychology.

Bailey raised the following objection to Berkeley's account: If (as Berkeley had admitted) neither visual nor tactile sensations alone can convey information about distance or "outness," then no mere association of such sensations can convey such information either. Bailey, a follower of Reid, claimed that we directly perceive distance visually. In response, Mill claimed that our perceptual judgments about distance involve a form of **ampliative inference** that goes beyond the information presented in visual and tactile sensation and accused Bailey of failing to distinguish between information derived from sensation and information derived from inference.

Bailey responded that we have no introspective awareness of any such inferential process:

> I cannot recognize in my experience such a process as the sensation of color suggesting an external thing. I directly and immediately see the colored external object.
>
> —(1855–1863, 2, p. 35)

He further claimed that we have no introspective awareness of the associated visual and tactual sensations from which judgments about distance are supposedly inferred:

> When I see an object under ordinary circumstances, I am not conscious of any affection in the organ of sight. I am conscious of perceiving the object at some distance but not of any sensation in the eye itself.
>
> —(1855–1863, 2, p. 40)

Mill granted both these points. However, he maintained that the visual perception of distance involves a form of unconscious inference or "unconscious cerebration" (Carpenter, 1874), based upon the association of visual and tactile sensations. Mill was the first to explicitly postulate a **rational unconscious**, governed by norms of rationality and logical inference (Reed, 1997). According to Mill, this unconscious inference is so automatic we naturally mistake it for a form of direct perception (1865, p. 166). Mill's account of perception influenced many later psychologists, notably Helmholtz and Wundt.

Mill was successful in defending Berkeley's account of distance perception, and Bailey's critique was quickly dismissed (Pastore, 1965). However, in defending Berkeley, Mill transformed associationist psychology almost beyond recognition, by sacrificing two fundamental principles of dogmatic British empiricism. In the first place, he abandoned the notion that scientific theories should be restricted to objects of conscious experience, by postulating that distance perception involves unconscious inference (Berkeley had rejected Descartes' theory of distance perception in terms of geometrical computations precisely because he had no conscious awareness of such computations). In the second place, Mill abandoned the notion that we have direct introspective access to all our mental states. He acknowledged that we do not have introspective access to the elemental visual and tactile sensations upon which our perceptual inferences are supposedly based: He noted that associated visual and tactile sensations become so integrated within perceptual judgment that the original sensations become "dim, confused, and difficult to be recalled" (1865, p. 180). Accordingly, later psychologists who developed Mill's account of perception as a form of unconscious inference from sensational elements increasingly relied upon physiological rather than introspective psychological data in support of their theories (Reed, 1997).

Alexander Bain: Psychology and Physiology

The last great British empiricist and associationist psychologist was Alexander Bain (1818–1903). His two-volume survey of contemporary associationist psychology and physiology, *The Senses and the Intellect* (1855) and *The Emotions and the Will* (1859), was the standard British text in psychology during the latter half of the 19th century and represented the first modern textbook of psychology.

Bain was the largely self-educated son of a poor weaver. He attended Marischal College, Aberdeen, where he attained top academic honors, and later traveled to London, where he befriended John Stuart Mill. The two met regularly to discuss their evolving ideas on philosophy and psychology, and Mill was so impressed by Bain that he asked him to read the proofs of *A System of Logic*. Mill in turn supported his protégé by persuading his own publisher to produce Bain's *The Senses and the Intellect.* When it lost money, Mill guaranteed the reluctant publisher 100 pounds sterling against losses on the second volume, *The Emotions and the Will*. Mill's praise for Bain's work in essays and reviews undoubtedly contributed to the eventual success of Bain's volumes. According to Mill, Bain was the first to achieve a substantive integration of psychology and physiology, based upon contemporary research in the physical sciences (Mill, 1859/1867).

Despite his best efforts, Bain originally failed to attain a university position and seemed fated to spend his years teaching geography at a finishing school for young women in London. He supplemented his meager income with royalties from articles in popular magazines such as the *Westminster Review* (on topics such as sympathy and toys) and from editorial work on the physiology of the nervous system. However, with the eventual success of his "big book," he was offered the Chair of Logic at Marischal College, Aberdeen, in 1860, where he remained until his retirement in 1876. That same year he founded the journal *Mind: A Quarterly Review of Psychology and Philosophy* with George Croom Robertson (1842–1892), who became the first editor. The first issues of the journal were devoted to the question of whether psychology could be a genuine science: The founding editorial hoped that the publication of the journal would enable its readers "to procure a decision on . . . the scientific standing of psychology" (Robertson, 1876, p. 3). Over time the focus shifted to purely philosophical questions, and *Mind* eventually became the premier British journal in philosophy, although reference to psychology in the subtitle was not dropped until 1974 (Neary, 2001).

The Senses and the Intellect and *The Emotions and the Will* were the first texts to integrate associationist psychology and the important developments of 19th-century physiology. They set the standard for later psychology texts, whose authors felt obliged to include some account of the structure and function of the nervous system. Although Bain maintained the traditional empiricist and associationist commitment to introspection, he acknowledged that "consciousness is not indispensable to the operations of intellect" (1855, p. 316). He also recognized the innate basis of many features of human and animal psychology and behavior, such as emotions and instincts.

In many respects Bain's texts were transitional. They looked backward to traditional associationist psychology and the recent history of physiology and forward to theories of evolution and late-19th-century advances in neurophysiology. Bain updated these texts through four editions, but they were frequently outdated by the time the latest edition was published. Bain's own psychological and physiological positions were largely secondhand. His psychological theories were not based upon extensive introspective or behavioral observation, and he did no clinical or physiological work. *The Senses and the Intellect* (1855) was published the same year as Herbert Spencer's *Principles of Psychology,* and *The Emotions and the Will* (1859) the same year as Charles Darwin's *On The Origin of Species*. Yet although Bain included sections on their theories of biological and mental evolution in later editions of his work, he made little attempt to integrate associationist psychology with evolutionary theory.

Bain advanced a fairly standard account of the association of ideas and behavior in terms of contiguity with repetition:

> Actions, sensations, and states of feeling, occurring together or in close succession, tend to grow together, or cohere in such a way that, when any one of them is afterwards presented to the mind, the others are apt to be brought up in idea.
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> —(1855, p. 318)

He followed John Stuart Mill in reintroducing similarity as a fundamental principle of association, which he believed was necessary to explain higher mental processes, notably those involved in analogical reasoning. Although he endorsed the basic principles of psychological and meaning empiricism, Bain was careful to point out that these principles are less restrictive than commonly supposed. He granted that our ideas are derived from sensory experience, but noted that it does not follow that we must have prior sensory experience of complex entities in order to form ideas or concepts of them. Bain emphasized the creative combinatory and developmental possibilities of ideation. He claimed that we are able to form meaningful ideas of possibly nonexistent entities via novel combinations of simple ideas, such as our idea of a "golden mountain," and via analogical extensions of complex ideas, such as our theoretical notion of "light waves" (Bain, 1855, p. 571).

Voluntary Behavior Bain's most fertile development of associationist psychology was his account of voluntary behavior, which he treated as a form of learned behavior based upon association. Unlike involuntary reflexive behavior, voluntary behavior is often generated *independently* of the stimulation of external sensory receptors.

Behavior can be generated independently of sensory stimulation, according to Bain, because nervous energy stored within the organism may be spontaneously discharged to motor nerves without antecedent stimulation ("where no stimulus from without is present as a cause"):

> There is in the constitution a store of nervous energy, accumulated during the nutrition and repose of the system, and proceeding into action with, or without, the application of outward stimulants.
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> —(1859, p. 328)

According to Bain, such spontaneously generated behavior is converted into directed or purposive voluntary behavior when it becomes associated with the experience of pleasure and pain, as in the case of a newborn lamb progressively coordinating originally spontaneous movements until they develop into purposive movements toward its mother's teat (1855, pp. 404–405).

This account of behavioral learning, according to which behaviors followed by success, satisfaction, or pleasure tend to be repeated, later became known as the **Spencer-Bain principle** (Boakes, 1984). Earlier versions of Bain's account of voluntary behavior are to be found in Hartley, Erasmus Darwin, and the German physiologist Johannes Müller (1801–1858), from whom Bain may have derived his account (Müller is cited extensively in Bain's discussion of voluntary behavior). Both Müller and Bain stressed that the associative processes that transform spontaneous activity into voluntary behavior are generally unconscious, since they operate in lower animals and neonates as well as in adult humans.

Bain's distinction between involuntary (reflexive) and voluntary behavior also anticipated later distinctions between responsive (stimulus determined) and operant (consequence determined) forms of conditioning, and Bain recognized instances of both. He cited a number of stimulus-determined associative reflexes that Pavlov later characterized as conditioned reflexes:

> The mere idea of a nauseous taste can excite the reality even to the point of vomiting. The sight of a person about to pass a sharp instrument over glass excites the well-known sensation in the teeth. The sight of food makes the saliva begin to flow.
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> —(1868, p. 90)

CEREBRAL LOCALIZATION

The 19th century saw great advances in neuroanatomy, especially during the 50-year period between Franz Joseph Gall's *On the Functions of the Brain* (Gall & Spurzheim, 1822–1825/1835) and David Ferrier's *The Functions of the Brain* (1876). This period also saw a marked shift of emphasis from correlational to controlled experimental studies of neurophysiological functions, a pattern later repeated in the development of comparative psychology and scientific psychology in general.

Despite advances in the neurophysiological location of psychological functions, the 19th century did not represent a progressive triumph of materialism and the reductive physiological explanation of human and animal psychology and behavior. On the contrary, many of the pioneers of neurophysiological localization either championed a form of substance dualism or maintained a neutral parallelism, which enabled them to avoid familiar charges of materialism, atheism, and fatalism. In the early part of the century at least, even those who abandoned substance dualism maintained a form of **neurophysiological dualism**, which preserved the rational autonomy of the human intellect and will championed by traditional substance dualists such as Descartes. Many early theorists held that the cognitive functions of the cerebral cortex are categorically distinct from the sensory-motor functions of the lower brain and spinal cord.

In the early 19th century the English physiologist Marshall Hall (1790–1857) established that there are numerous connections between sensory and motor nerves in the spinal cord and introduced the notion of a **reflex arc**: a system comprising a sensory nerve, interconnecting nervous tissue in the spinal cord, and a motor nerve (Boakes, 1984). Hall distinguished between the "excitory-motor" system, which he located in the lower brain and spinal cord (the "true spinal" system), and the "sensory-volitional" system, which he located in the cerebral cortex. According to Hall, the reflexive excitory-motor system accounts for automatic, instinctual, and emotional behavior, whereas the sensory-volitional system accounts for rational, learned, and purposive behavior.

Franz Joseph Gall: Phrenology

Franz Joseph Gall (1758–1828), a Viennese physician and anatomist, developed what became known as **phrenology**, the doctrine that the degree of development of psychological faculties is a function of the size of the area of the brain in which they are localized, which can be determined by measurements of the contours of the skull, or **cranioscopy**. Gall tried to map the functions of the brain by establishing correlations between behavioral manifestations of psychological faculties and protrusions and indentations of the skull, supposedly caused by the development or underdevelopment of the associated "separate organs" of the brain. According to Gall, a developed faculty of acquisitiveness, for example, is reflected in a protrusion just above and in front of the left ear; an underdeveloped faculty of acquisitiveness is marked by an indentation in the same place. Human behavior can be explained and predicted by reference to the degree of development of the contours, or "bumps," on the skull.

Gall's theory was reputedly inspired by his childhood observation that classmates who excelled in rote memory had "large prominent eyes" and his belief that such correlation was not accidental (Young, 1990). His medical training led him to the conclusion that "the difference in the form of heads is occasioned by the difference in the form of the brains" (Gall & Spurzheim, 1822–1825/1835, 1, p. 59). Gall claimed that moral and intellectual faculties are innately determined, in contrast to the optimistic environmentalism of French sensationalists such as Condillac and idéologues such as de Tracy. He argued that individual differences in psychological faculties and propensities among humans, and between humans and animals, cannot be explained in terms of environment and learning history, but must be explained in terms of biological endowment.

Such postulated limits on human intellectual and moral perfectibility led to inevitable charges of materialism, atheism, and fatalism, although Gall declined to take any position on the mind-body problem. He was forced to leave Vienna and move to Paris when the Catholic Church and the Austrian authorities condemned his works. They were placed on the Index of Prohibited Books, and Gall was denied a Christian burial when he died in 1828. However, his doctrine attracted many followers, notably Johann Casper Spurzheim (1776–1832), who collaborated with Gall on the publication of *The Anatomy and Physiology of the Nervous System* (in four volumes between 1810 and 1819, closely followed by popular editions of the same text). It was Spurzheim who coined the term *phrenology,* which Gall never used (Clarke & Jacyna, 1987).

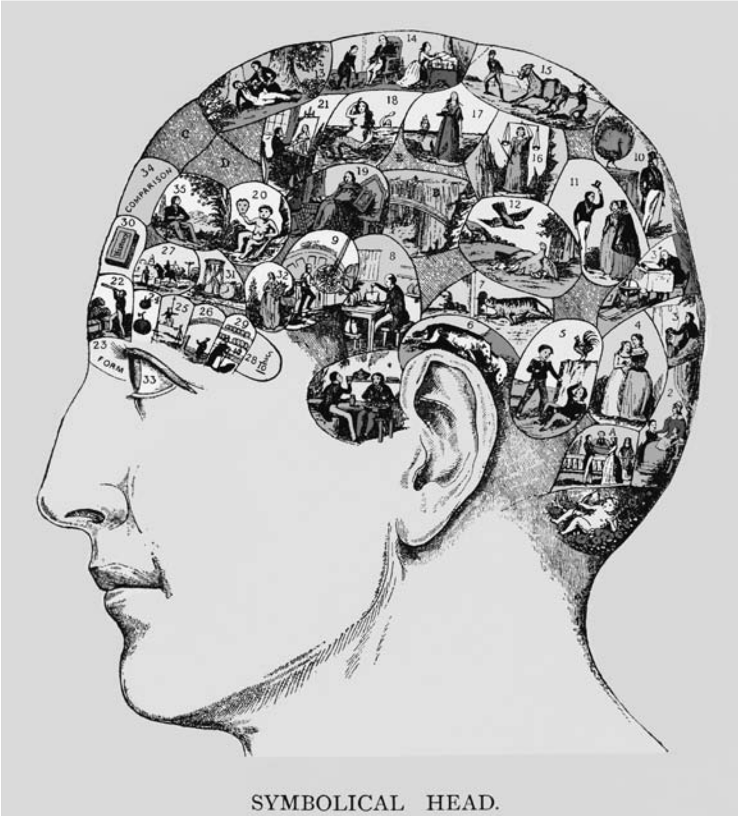
Empirical and Biological Psychology The scientific community originally treated Gall's work with respect. He was a skilled neuroanatomist whose dissection techniques were much admired, since they represented a significant improvement over traditional "mutilative" techniques. Gall was largely responsible for developing the surgical methods that enabled experimental physiologists to leave discrete convolutions of the brain intact (O'Donnell, 1985). However, his specific claims about the neural localization of particular psychological faculties and propensities were undermined by later research (which, ironically, employed the very same surgical methods that Gall had pioneered). Gall and his followers were also overly enthusiastic in their appeal to positive instances of correlation between protrusions of the skull and behavioral manifestations of psychological faculties and uncritically dismissive of negative instances in which no correlation was found. Their attempts to explain away negative instances by appealing to brain disease or damage or by withdrawing original attributions of a developed faculty (when Descartes' skull was found to lack the relevant protrusion for rationality, they concluded that Descartes had not been as great a thinker as had been previously supposed) led to the justified dismissal of phrenology as a pseudoscience, on a par with palmistry and astrology.

This was unfortunate, since Gall's attempt to develop an empirical biological psychology presaged a number of later developments in physiology and psychology. Although medieval "inner sense" theorists had speculated about the ventrical location of psychological faculties such as cognition and memory, Gall was the first to attempt to empirically identify the neural location of specific faculties. His localization of psychological faculties was based upon his study of the skulls of normal and abnormal adults, children, and the elderly, and his comparative analysis of the psychological faculties of different species of animals and men (even if he relied too much on anecdotal reports and was too cavalier in his dismissal of negative findings). Gall was arguably the first empirical physiological psychologist, even though later experimental researchers came to disparage his naturalistic correlational methods.

Gall maintained that the "fundamental, primitive faculties" of animals and humans should be established empirically. He was critical of the types of faculties postulated by empiricists and sensationalists, who focused almost exclusively on epistemological faculties such as perception, cognition, and memory. In contrast, Gall focused on adaptive and socially oriented faculties such as the "carnivorous instinct," the "maternal instinct," and the "disposition to murder," in addition to traditional cognitive and moral faculties (Young, 1990).

Gall insisted that anyone concerned with the objective study of the neurophysiological basis of psychological functions "should have a clearly defined conception of what he is looking for" (1822–1825/1835, 3, p. 160). According to Gall, psychological functions can be established only via the comparative study of the behavioral repertoires of normal adult humans, children, animals, and the insane. He insisted that only after empirical categories of psychological functionality have been established are neurophysiologists in the position to systematically correlate psychological functions with neurophysiological locations. Unfortunately, Gall's prescription was neglected by later generations of neurophysiologists and, to the detriment of his own legacy, often enough by Gall himself, who adopted many of the traditional categories of Scottish common sense psychology, such as self-preservation, duty, love, and imitation, not to mention "the instinct for property owning and stocking up on food."

Gall identified 27 fundamental faculties, atomistically conceived as distinct and independent. Gall claimed that animals share 15 of these with humans (Young, 1990), but did not endorse strong continuity between human and animal psychology and behavior: He maintained that there are 12 human faculties that animals do not have to any degree. Gall worked in a pre-evolutionary period and believed in a fixed natural hierarchy. However, his comparative studies of psychological faculties in different species, his emphasis on behavior and its adaptive function, and his stress on variation between and within species presaged later developments in comparative, functional, and differential psychology, although his commitment to the pseudoscience of phrenology relegated his own legacy to the intellectual dustbin of history.



Phrenological head representing psychological faculties.

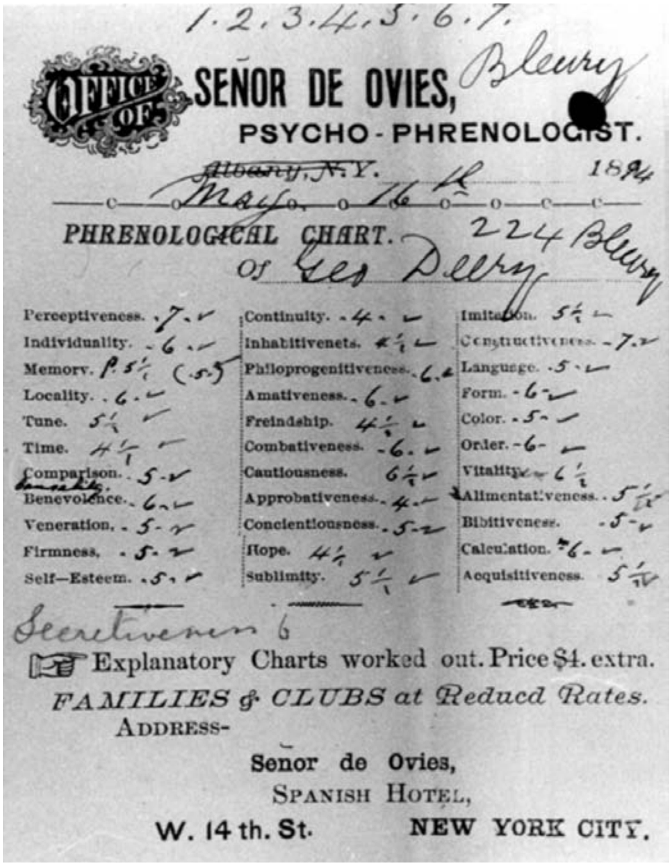
While empirically discredited, Gall's psychology, with its emphasis on individual differences, anticipated the forms of functional and behaviorist psychology that dominated American psychology in the early half of the 20th century. William McDougall (1871–1938) employed Gall's methods of behavioral analysis of psychological functions in his influential 1908 work on instinct, *Introduction to Social Psychology*. Like Comte and later behaviorist psychologists, Gall was opposed to introspective psychology. He believed that introspective methods distorted psychological investigation in much the same way as traditional dissection techniques distorted neurophysiological investigation and that they posed a major threat to the development of an objective psychological science.

After Gall died in 1828, Spurzheim and his Scottish disciple George Combe (1788–1858) promoted phrenology in Europe and America. Sales of Combe's 1827 text *Essay on the Constitution of Man and Its Relation to External Objects* reached six figures, and Spurzheim toured America to great acclaim in 1832 (Walsh, 1972). Combe's American lectures of 1838–1840 were attended by physicians, ministers, educators, asylum superintendents, and college professors, who saw phrenology as a source of potentially useful knowledge (O'Donnell, 1985).

Phrenological societies and consulting offices were founded in major European and American cities. In the hands of American entrepreneurs such as Orson Fowler (1809–1887), Lorenzo Fowler (1811–1896), and Samuel Wells (1820–1875), who developed elaborately labeled busts and manuals for self-analysis, phrenology became big business. It eventually took on the status of a cult rather than a scientific discipline, which explains why it survived long after empirical demonstrations of its inadequacy and retained adherents even in the late 20th century (Leek, 1970).

Applied Phrenology Phrenology was especially popular in America in the years prior to the Civil War, when itinerant phrenologists gave public lectures and demonstrations in churches and town halls and did private "delineations" to paying clients. The *Annals of Phrenology* began publication in 1833; the *American Phrenological Journal and Miscellany* began publication in 1838 and ran until 1911. Phrenologists offered vocational guidance to paying clients, and in some places potential employers required phrenological analyses. Sizer (1882) claimed that those with highly developed faculties of acquisitiveness and secretiveness were especially suited to be merchants and bankers (cited in Sokal, 2001), and railway companies considered using phrenologists to select competent trainmen in order to reduce accidents (O'Donnell, 1985). Phrenologists also offered child-rearing advice and marriage counseling. They advised prospective husbands and wives to marry those with underdeveloped faculties that matched their own developed faculties (and vice versa). Many flocked to the New York parlors of Fowler and Wells to receive swift diagnoses of their vocational aptitudes and marriage prospects (O'Donnell, 1985).

In their practical applications, American phrenologists went considerably beyond Gall's original theory. In contrast to Gall's commitment to the innate biological determination of human psychological faculties, American phrenologists stressed their plasticity and counseled their clients to cultivate or constrain them. The popular reception and real if limited efficacy of phrenological counseling is probably best explained in terms of placebo effects and the commonsensical nature of the advice offered, which was rarely based upon the specifics of phrenological theory. In this respect, the practice of early American phrenologists anticipated that of early-20th-century American applied psychologists, whose pioneering explorations in educational, industrial, and clinical psychology were often based more on common sense and practical experience than on the theoretical systems of scientific psychology (Bakan, 1966). Whatever their actual degree of success, the professional practice of American phrenologists nourished public expectations for a scientifically objective but socially useful form of psychology of the type later promoted by American functional and behaviorist psychologists (O'Donnell, 1985).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Pierre Flourens: Experimental Physiology

The French surgeon Marie-Jean-Pierre Flourens (1794–1867) undertook the first systematic critique of phrenology in *An Examination of Phrenology* (1843) and *On Phrenology* (1863). Flourens established that many of Gall's localizations of psychological faculties were inaccurate and undermined the foundation of phrenology by demonstrating that the skull does not reflect the contours of the brain. In contrast to Gall, Flourens was a respected figure of the French scientific establishment. He received the Montyon prize in experimental physiology in 1824 and 1825 and shortly after was elected to the French Academy of Sciences (an honor refused to Gall). Flourens's pioneering experimental studies of the nervous system were documented in *Experimental Research on the Properties and Functions of the Nervous System in Vertebrates* (1824). He became professor of comparative physiology at the Collège de France in 1832 and received the ribbon of the Legion of Honor the same year (Young, 1990).

Experimental Ablation Flourens's experimental studies were a continuation of the program of neural localization initiated by Albrecht von Haller (1708–1777). Von Haller, who set the agenda for modern physiology in *Elements of Physiology* (1757–1765), pioneered the use of experimentation on live animals to identify physiological functions. Flourens extended this methodology to the exploration of the vertebrate nervous system and perfected the experimental method of **ablation**. This involves the systematic removal of neural tissue from live animals and careful observation of their consequent behavior in order to determine the function of the extirpated part of the nervous system. Although Flourens was not the first to employ the method of ablation, his meticulous surgical treatments set the standard for future research (later supplemented by electrical and chemical stimulation and electronic recording and graphing). Physiologists came to treat experimental intervention as the mark of a scientific approach to neural localization and rejected the correlational methods of Gall and his followers.

Flourens was the first to employ ablation to determine the functions of neural structures. His enduring achievement was to establish the cerebellum as the center for the control of motor behavior and the medula oblongata as the center for the control of vital functions such as respiration and heartbeat. He also extended von Haller's work on the **irritability**, or "excitability," of nerves, conceived of as the ability to transmit excitation that results in muscle contraction or sensation. On the basis of a series of experimental studies, Flourens concluded that irritability is not a universal property of the nervous system, since he found no evidence of it in the cerebral cortex.

As a means of localization of neural functions, Flourens's controlled ablative studies were in many respects superior to Gall's naturalistic correlations between behavior and protrusions and indentations of the skull. Yet he uncritically dismissed Gall's novel attempt to empirically establish a classification of psychological functions based upon distinctive behavioral repertoires (Young, 1990). Gall complained with some justification that the work of "mutilators" such as Flourens had little value, since it was not based upon empirical knowledge of "fundamental powers."

The Functional Unity of the Cerebral Cortex Although his stated aim was to "ascertain experimentally . . . which parts of the nervous system are used exclusively for sensation, which for [muscle] contraction, which for perception, etc." (1842, p. 3), Flourens was a very limited champion of neural localization. For his opposition to Gall went beyond disagreement about the neural location of particular psychological faculties and the advocacy of experimental over correlational methods. Flourens claimed that the cerebral cortex is the center of perception, will, and intelligence, but denied that these traditional faculties (and associated faculties of reasoning, memory, and judgment) are located in specific regions of the cerebral cortex. His denial was partly based upon the apparent lack of irritability of the cerebral cortex but also upon his conviction that the cognitive functions of the cerebral cortex are unitary. According to Flourens, the diverse functions of the lower nervous system, including the sensory-motor system localized in the lower brain and spinal cord, are presided over by the unitary cognitive functions of the cerebral cortex.

Flourens's claim that the integrated functions of perception, will, and intelligence are subserved by the "whole mass" of the cerebral hemispheres anticipated later doctrines of mass action and equipotentiality (Lashley, 1929). However, Flourens's primary ground for this claim was his commitment to substance dualism. According to Flourens, the unified cognitive functions of the cerebral cortex are the expression of the unified powers of the immaterial soul. This constituted his fundamental objection to Gall's project of neural localization: Gall had denied the unity or "indivisibility" of the immaterial soul that Descartes had affirmed. Like Descartes, to whom he dedicated his *Examination of Phrenology,* Flourens believed that the denial of the unity of the immaterial soul would lead to materialism, atheism, and fatalism.

François Magendie: The Bell-Magendie Law

The French physician François Magendie (1783–1855) abandoned his anatomical studies and surgery to focus on experimental physiology in 1813. He developed the first course on physiology as an autonomous discipline and founded the *Journal de Physiologie Expérimentale* (*Journal of Experimental Physiology*) in 1821. Magendie sought to establish physiology as a natural science in much the same fashion that Galileo and Newton had previously established physics and astronomy as natural sciences. He described this as the aim of his physiological textbook, *An Elementary Treatise on Human Physiology* (1838/1843), in which he championed experimentation as the basis of physiological and medical knowledge. Magendie was also a powerful member of the French scientific establishment, whose work inspired later generations of experimental physiologists, notably Claude Bernard (1813–1878) and Louis Pasteur (1822–1895), who were instrumental in establishing the supremacy of experimental over correlational methods in physiology.

Sensory and Motor Nerves Magendie's major contribution to experimental physiology was his demonstration of the location of distinct sensory and motor nerves in the spinal cord. Hartley had distinguished between sensory and motor nerve pathways, and von Haller and Whytt had located sensory-motor reflexes in the spinal cord; but Magendie was the first to demonstrate the separate locations of sensory and motor nerves in the posterior and anterior roots of the peripheral nerves in the spinal cord (respectively). Magendie exposed the spinal cord of a six-week-old puppy and found that severance of the anterior roots eliminated motor movements but left sensitivity intact and that severance of the posterior roots eliminated sensitivity but left motor movements intact. He concluded that

> the anterior and posterior roots of the nerves which arise from the spinal cord have different functions, that the posterior roots seem to be particularly destined for sensibility, while the anterior roots seem to be especially connected with movement.
>
> —(1822/1944, pp 101–102)

This was not an original discovery. The British physician and anatomist Charles Bell (1774–1842) had reported a similar result in 1811. He had exposed the spinal cord of stunned rabbits and noted how stimulation of the anterior roots produces convulsive movements, but stimulation of the posterior roots does not (Boakes, 1984). Bell believed he had identified the posterior and anterior roots as the vehicles of sensibility and movement, but his report was only circulated privately among friends. Knowledge of Bell's work led to a dispute about who deserved credit for the discovery of what later became known as the **Bell-Magendie law**. Flourens gave primary credit to Bell, possibly because of his own friction with Magendie, who was unpopular with his colleagues because of his vanity, jealousy, and fiery temper (Young, 1990). Bell was the first to attribute sensory and motor functions to the posterior and anterior spinal roots, but Magendie was the first to provide complete experimental confirmation of the different functions.

Cognition and Sensory-Motor Function Magendie hoped to extend his experimental methods to the exploration of the higher cognitive functions of the cerebral cortex, but did little to illuminate their nature. He followed Flourens in categorically distinguishing the cognitive functions of the cerebral cortex from the sensory-motor functions of the lower brain and spinal cord. He avowed that cognitive functions could be studied as "the result of the action of the brain," but also held that they may be "dependent upon the soul" (1838/1843, p. 146). He was able to maintain this position because he endorsed a neutral parallelism and adopted a positivist attitude to the study of the physiological correlates of mental states. He claimed that science can only describe the correlation between mental and physiological states, but cannot causally explain the relation between them.

Johannes Müller, who conducted a parallel series of experiments on frogs, confirmed the Bell-Magendie law. Like Magendie, he agreed with Flourens that the cerebral cortex is not irritable and that the unitary cognitive functions are not localized to specific regions of the cortex. According to Müller, the various cognitive functions of the cerebral cortex (such as perception, thought, and memory) are merely "different modes of action of the same power" (1833–1840/1838, 1842, p. 1345).

Müller characterized the relation between the higher cognitive functions of the cerebral cortex and the sensory-motor functions of the lower brain and spinal cord in terms of a famous metaphor. He claimed that the will acts upon the lower brain centers like a musician playing on the keyboard of a pianoforte:

> The fibres of all the motor, cerebral and spinal nerves may be imagined as spread out in the medula oblongata, and exposed to the influence of the will like the keys of a pianoforte.
>
> —(1833–1840/1838, 1842, p. 934)

Physiologists in the first half of the 19th century generally accepted this characterization. It represented a form of neurophysiological dualism that preserved the autonomy of human thought, rationality, and will avowed by traditional dualists such as Descartes.

Pierre-Paul Broca: Aphasia

The French physician Pierre-Paul Broca (1824–1880) is often credited as the first person to have identified a specific neural location associated with a distinctive psychological function. Broca localized the "faculty of articulate language" to the superior region of the left frontal lobe, now known as **Broca's area**, based upon an autopsy performed upon his patient "Tan," who died within a week of admission to Broca's surgery (Broca, 1861/1960). Tan (whose real name was Leborgne) had lost his speech 21 years earlier and had been a patient at La Bicêtre Hospital in Paris.

Broca was neither the first person to study aphasia, nor the first to relate it to brain damage. Speculation about speech pathology and its neural origin goes back to the ancient Greeks, and Gall was the first to offer a "complete description of aphasia due to a wound in the brain" (Head, 1926, I, 9, cited in Young, 1990, p. 135). However, Broca gathered his evidence at a time when the scientific community was prepared to take it seriously. He presented his results in the course of an academic controversy over whether the cognitive functions of the cerebral cortex are unitary, as Flourens had maintained, or discrete, as Gall had maintained. Broca was secretary to the newly founded *Société d'Anthropologie*, where the debate had recently focused on the functions of a primitive skull presented by one of the members.

On the basis of the case of Tan and other autopsies, Broca claimed that he had isolated an autonomous faculty of language, which could be eliminated without damage to other intellectual faculties (such as memory and intelligence) and thus refuted Flourens's contention that the cerebral cortex acts "as a whole." However, some were cautious about Broca's evidence based upon "facts furnished by the experiments of disease in man" (Ferrier, 1876/1886, p. 270), especially since some of Broca's patients suffered from extensive brain damage and atrophy (Young, 1990).

Although Broca located the faculty of articulate language in the cerebral cortex, he followed Flourens and Magendie in maintaining that the functions of the cerebral cortex are essentially cognitive. He maintained the traditional distinction between the higher intellectual functions, attributed to the cerebral cortex, and the lower sensory-motor functions, attributed to the lower brain and spinal cord.

Gustav Fritsch and Eduard Hitzig: The Excitability of the Cerebral Cortex

Much of the early-19th-century debate about the cerebral localization of psychological functions took place in France, but it later shifted to Germany and Britain. The traditional distinction between the cognitive functions of the cerebral cortex and the sensory-motor functions of the lower brain and spinal cord had been supported by the experimentally demonstrated excitability of the nerves of the lower brain and spinal cord and the generally acknowledged inexcitability of the cerebral cortex. The work of Gustav Fritsch (1839–1927) and Eduard Hitzig (1838–1907) undermined this traditional distinction.

In a series of studies conducted on dogs and rabbits on a dressing table in a small Berlin house, Fritsch and Hitzig demonstrated that the cerebral cortex responds to electrical stimulation and that one region of the cortex is responsible for muscular contractions. They published their results in a paper titled “On the Electrical Excitability of the Cerebrum” (1870), whose significance was immediately recognized by the scientific community.

Like Broca, they rejected Flourens’s view that the cerebral cortex acts “as a whole.” However, they shared his commitment to dualism and maintained that the immaterial soul acts through the different regions of the cerebral cortex with their localized functions:

> It further appears, from the sum of all our experiments that the soul is not, as Flourens and others after him had thought, a function of the whole of the hemispheres, the expression of which one might destroy by mechanical means in the whole, but not in its various parts, but that on the contrary, certainly some psychological functions and perhaps all of them, in order to enter matter or originate from it need certain circumscribed centers of the cortex.
>
> —(1870, p. 96)

Their experiments were quickly replicated, initially by David Ferrier (1843–1928) in Britain (Ferrier, 1873) and by Leonardo Bianchi (1848–1927) in Italy (Bianchi, 1895). Ferrier, who was chair of forensic medicine at Kings College, London, mapped the motor cortex via a series of experiments on dogs, cats, rabbits, and guinea pigs. His 1876 book *The Functions of the Brain* (1876) represented the triumph of neural localization and initiated a program of research in England, Germany, France, and

Italy directed to the detailed mapping of the discrete functional centers of the cerebral cortex. Experimental explorations of the cortex, facilitated by improved techniques of ablation and stimulation, led to the identification of localized areas for different motor movements and the identification of the sensory centers for vision, hearing, touch, and other sensory modalities. This "new phrenology" had one immediate practical benefit: It greatly advanced the prospects of effective neural surgery by establishing targeted sites for the removal of suspected tumors or swelling (Young, 1990).

These pioneering experiments were also replicated on Mary Rafferty, an Irish domestic servant who was admitted to the Good Samaritan Hospital in Cincinnati in 1874 with a cancerous ulcer on the side of her head that exposed her brain through a 2-inch hole in her skull. One of her attending physicians, Robert Bartholow (1831–1904) of the Medical College of Ohio in Cincinnati, decided to exploit this opportunity to replicate the studies of Fritsch and Hitzig and Ferrier. He passed an electric current through needles inserted into her brain, which produced convulsive movements analogous to those of laboratory animals. In a paper titled "Experimental Investigations Into the Functions of the Human Brain" (1874), Bartholow described the effects on electrical neural stimulation on his unfortunate patient:

> When the needle entered the brain substance, she complained of acute pain in the neck. In order to develop a more decided reaction, the strength of the current was increased by drawing out the wooden cylinder one inch. When communication was made with the needles, her countenance exhibited great distress, and she began to cry. Very soon, the left hand was extended as if in the act of taking hold of some object in front of her; the arm presently was agitated with clonic spasm; her eyes became fixed with pupils widely dilated; her lips were blue, and she frothed at the mouth; her breathing became stertorous; she lost consciousness and was violently convulsed on the left side. The convulsion lasted five minutes, and was succeeded by a coma. She returned to consciousness in twenty minutes from the beginning of the attack, and complained of some weakness and vertigo.
>
> —(1874, pp. 310–311)

After the experiments, Mary began to complain of headaches and died a few days later; her death was certified as due to cancer.

The American Medical Association condemned Bartholow, although he insisted that he had performed the experiment with Mary's consent and maintained that it did not cause her death. He was forced to resign his professorship at the Medical College of Ohio, but went on to have a distinguished career as professor of medicine and dean of Jefferson Medical College in Philadelphia (Lederer, 1995). Bartholow published a number of successful books on medicine and therapeutics. He became a fellow of the College of Physicians of Philadelphia and a member of the American Philosophical Society. He was later elected president of the American Neurological Association.

The Sensory-Motor Theory of the Nervous System

The cerebral localization of centers for sensation and movement promoted the development of the **sensory-motor theory** of the nervous system, according to which the whole nervous system is a reflexive sensory-motor system, whose every component can be characterized as having a sensory or motor function (Danziger, 1982). On this theory, the higher cognitive functions of the cerebral cortex are strongly continuous with the lower sensory-motor functions of the lower brain and spinal cord: They are merely more complex elaborations of the reflexive sensory-motor functions to be found in humans and animals.

Thomas Laycock (1812–1876), professor of medicine at the University of Edinburgh, was one of the first to maintain that the principles that govern the reflexive system of the lower brain and spinal cord should be extended to the cerebral cortex:

> The brain, although the organ of consciousness, is subject to the laws of reflex action, and . . . , in this respect, it does not differ from the other ganglia of the nervous system. . . . The ganglia within the cranium, being a continuation of the spinal cord, must necessarily be regulated as to their reaction on external agencies by laws identical to those governing the functions of the spinal ganglia and their analogues in the lower animals.
>
> —(1845, p. 298)

John Hughlings Jackson, a former student of Laycock, treated sensation and movement as the basic elements of human psychology and claimed that they are instantiated in the cerebral cortex as "nervous arrangements representing impressions and movements" (1931, 1, p. 42). He argued that reflexive sensory-motor functions previously attributed exclusively to the lower brain and spinal cord should be attributed to the cerebral cortex, since he maintained that reflexive "sensori-motor processes are the physical side of, or . . . form the anatomical substrata of, mental states" (1931, 1, p. 49).

Jackson supported his sensory-motor theory with clinical autopsies of aphasics and epileptics, which revealed various forms of disease of or damage to the cerebral cortex. He claimed that all mental disorders caused by disease of or damage to the cerebral cortex, such as aphasia, epilepsy, and delirium are due to "lack, or to disorderly development, of sensori-motor processes" (1931, 1, p. 26). In contrast to Broca, who had claimed that aphasia is a cognitive disorder, Jackson maintained that it is a motor disorder, a defect of "articulatory movements" (Young, 1990). David Ferrier, who sought the "artificial reproduction of the clinical experiments produced by disease" (1873, p. 30), managed to create the convulsions of epilepsy by direct stimulation of the brain.

The identification of cortical centers for sensory and motor processes did not mandate the extension of reflexive sensory-motor forms of explanation to the cognitive operations of the cerebral cortex or the strong continuity of cognitive and sensory-motor functions presupposed by the sensory-motor theory of the nervous system. The location of sensory and motor centers in the cortex was consistent with the existence of distinct cognitive centers. Fritsch and Hitzig claimed that the anterior frontal regions of the cortex are responsible for abstract thought and play a minimal role in sensory-motor function, and Ferrier maintained that they are responsible for focused attention. Although the mechanistic explanation of all animal and human behavior undermined Descartes' account of voluntary behavior in terms of the free action of an immaterial soul, it was consistent with Bain and Muller's nonreflexive explanation of voluntary behavior as learned behavior generated independently of sensory stimulation.

However, many 19th-century physiologists embraced the reflexive sensory-motor theory of the nervous system and recognized the challenge it posed to traditional conceptions of mind, consciousness, and behavior. As George Croom Robertson, the editor of Bain's journal *Mind,* put it, 19th-century neurophysiology established a body of experimental results "to be reckoned with, by psychologists as well as physiologists" (Robertson, 1877, p. 92). Many 19th-century physiologists and medical psychologists extended reflexive explanation to cover the cognitive operations of the cerebral cortex, such as perception, memory, decision making, problem solving, and purposive behavior and focused their research on unconscious and automatic forms of cognition and behavior, such as hypnotic suggestion and somnambulism. For example, Laycock advanced reflexive explanations (albeit largely speculative) of complex, purposive but automatic behavior such as hysteria, impulsive insanity, and bizarre religious behavior in terms of cerebral reflexes (Danziger, 1982).

Ideomotor Behavior William Carpenter (1813–1885), professor of physiology at University College, London, was the author of the influential textbook *Principles of Human Physiology* (1855), which helped to establish physiology as an autonomous discipline in Britain. He followed Hartley and Bain in maintaining that the laws of association connect ideas with behavior as well as with other ideas. According to Carpenter, ideas associated with behavior, or ideas of behavior, can come to generate behavior as a consequence of association. He identified a class of automatic but apparently purposive behavior mediated by ideas that he called **ideomotor behavior** (Carpenter, 1874) and appealed to association to explain the efficacy of hypnoses and other forms of suggestion, in which ideas automatically produce behavior without the agent willing the behavior or being conscious of the connection between the idea and behavior. Carpenter also introduced the notion of **unconscious cerebration** (which Mill employed in his account of unconscious inference in perception) to explain involuntary attention, unconscious problem solving, and dreams and hallucinations (Danziger, 1982). His theory of suggestibility provided the plot of Wilkie Collins's *The Moonstone,* the first British detective novel (Reed, 1997).

Although he claimed that some human behavior is a product of "self-regulation" by the will, Carpenter insisted that the action of the will is entirely dependent upon the reflexive mechanisms governing ideomotor behavior. For Carpenter, the will never initiates behavior directly, but can act only by "direction of the attention" (1874, p. 25), by focusing on ideas that automatically generate behavior. Attention strengthens certain ideas at the expense of others, and enables individuals to determine which automatic behavior comes into play. In contrast, inattention or misdirected attention (to imagined debauchery, for example) can result in bad habits and dissolution. In his own medical practice Carpenter recommended that peculiar mix of moral exhortation and directive rote learning characteristic of Victorian morality. He urged his patients to focus their attention on pledges to avoid strong drink, drugs, and prostitution and to ingrain good conduct through repetition of socially accepted behaviors until they became habitual (or "secondary automatic," as Hartley had put it).

Epiphenomenalism One commonly represented consequence of the sensory-motor theory of the nervous system was the view that mentality and consciousness are merely epiphenomenal by-products of the reflexive mechanisms of the nervous system and play no role in the generation of behavior. Many theorists committed to the sensory-motor theory come to conceive of mentality and consciousness as "coincident" or "collateral" properties of those neurophysiological states that are responsible for the reception of sensory stimulation and the generation of behavior (Danziger, 1982).

The biologist Thomas Huxley championed this conception of mentality and consciousness in his address to the 1874 meeting of the British Association in Belfast, titled "On the Hypotheses That Animals Are Automata, and Its History." Huxley claimed that Descartes had been wrong to deny consciousness to animals, since the areas of the cerebral cortex established as the centers of consciousness in humans could be re-identified in animals such as apes and dogs, but maintained that 19th-century advances in neurophysiology supported Descartes' treatment of animals as automata. He claimed that the attribution of consciousness to animals is not inconsistent with their treatment as automata, because their consciousness is causally impotent with respect to their behavior:

> The consciousness of brutes would appear to be related to the mechanism of their body simply as a collateral product of its working, and to be as completely without any

> power of modifying that working as the steam-whistle which accompanies the work of a locomotive engine is without influence on its machinery. Their volition, if they have any, is an emotion indicative of physical changes, not the causes of such changes.
>
> —(1874, p. 575)

Huxley maintained that the same was true of humans, who, like animals, he characterized as **conscious automata**:

> The argumentation which applies to brutes holds equally good for men; and, therefore . . . all states of consciousness in us, as in them, are immediately caused by changes in the brain substance. . . . it follows that our mental conditions are simply the symbols in consciousness of the changes which take place automatically in the organism; and that, to take an extreme illustration, the feeling we call volition is not a voluntary act, but the symbol of that state of the brain which is the immediate cause of the act.
>
> —(1874, p. 577)

This doctrine came to be known as **epiphenomenalism**, although Huxley never used the term, which was originally employed to characterize symptoms of a disease that play no causal role in the progress of the disease. William James was the first to use the term to characterize Huxley's "conscious automaton-theory" (James, 1890, p. 129).

In defense of his claims about the causal impotency of mentality and consciousness, Huxley cited cases of animals and humans that engage in coordinated and purposive behavior despite decortication or temporary lack of consciousness due to brain damage. He noted that frogs with their cortex removed will continue to engage in such behavior, despite their presumed lack of consciousness: They will use their legs and feet to try to remove chemical irritants applied to their bodies (a phenomenon documented by Hales and Whytt in the 18th century). Huxley also described the case of a French army sergeant with a head wound, who suffered temporary periods of loss of consciousness, during which he would continue to eat, drink, smoke, and walk in the garden, while seemingly insensitive to pain and visual stimulation.

Yet at most these cases demonstrated only that animals and humans are able to engage in purposive behavior in the absence of consciousness, not that animal and human consciousness is impotent when it is present, far less that this is the case with respect to mentality per se (conscious or unconscious). As Huxley himself noted, a frog with a cortex responds to sights and sounds that the decorticated frog ignores, and the sergeant normally refused the quinine or vinegar that he happily drank during his periods of unconsciousness. Indeed, in a reference that would have delighted Descartes, Huxley noted that the sergeant was normally truthful and honest, but was a liar and a cheat during his unconscious lapses (1874, p. 572).

Epiphenomenalism was not mandated by either materialism or the sensory-motor theory of the nervous system, and neither La Mettrie nor Carpenter embraced the doctrine. Many psychologists continued to insist on the causal role of mentality and consciousness in the generation and control of behavior, although epiphenomenalism later found its most forceful expression in early American behaviorist psychology.

Control and Inhibition As the 19th century developed, neurophysiologists came to conceive of reflexive behavior in increasingly complex terms, as integrated adaptive reactions rather than isolated neuromuscular responses. Although the neurophysiological dualism of higher autonomous and lower reflexive processes was abandoned, the sensory-motor theory of the nervous system still retained elements of the traditional conception. The notion of distinct functional systems was replaced by the notion of a hierarchy of increasingly more complex reflexive systems, in which higher cerebral reflexes regulate the reflexes of the lower brain and spinal cord. Neurophysiologists also came to acknowledge that many reflexive behaviors are initiated centrally rather than peripherally, as the notion of the control of behavior by an immaterial will or autonomous cognitive center was replaced by the notion of control through cerebral inhibition. Ferrier treated reflexive cortical inhibition as the basis of a redefined notion of "voluntary" behavior:

> The primordial elements of . . . volitional acts . . . are capable of being reduced in ultimate physiological analysis to reaction between the centers of sensation and those of motion.
>
> But besides the power to act in response to feelings and desires, there is also the power to inhibit and restrain action, notwithstanding the tendency of feelings and desires to manifest themselves in active motor outbursts.
>
> —(1876, p. 282)

Whytt had demonstrated the enhancement of reflexes following decortication, and the liberating effects of drugs and alcohol on the cerebral cortex were well known. It became natural to offer explanations of epilepsy, aphasia, somnambulism, suggestibility, alcoholism, and insanity in terms of the breakdown of inhibitory cortical control of lower reflexive responses, through disease, damage, or the influence of chemical agents such as drugs or alcohol.

EXPERIMENTAL PHYSIOLOGY IN GERMANY

Given the major contributions to psychology by Hume, Hartley, Mill, and Bain, and to neurophysiology by Jackson, Carpenter, and Ferrier, one might have expected that institutional scientific psychology would have naturally developed in Britain at the end of the 19th century. After all, Bain's *Senses and the Intellect* (1855) predated Wundt's *Principles of Physiological Psychology* (1873–1874) by 18 years, and his journal *Mind* (1876–) predated Wundt's journal *Philosophical Studies* (1881–) by five years.

However, academic psychology developed much later in Britain than in the rest of Europe and America. In 1877 James Ward and John Venn tried to get psychology introduced as an academic discipline at the University of Cambridge, but the University Senate rejected their proposal. Ward did manage to get a grant of 50 pounds for psychological equipment in 1891 and secured a lectureship in experimental psychology and physiology of the senses in 1897. But psychology never really developed as an autonomous discipline in Britain until after the First World War, and many universities remained unreceptive until after the Second World War (Hearnshaw, 1964).

This was due in part to the inherent conservatism of the British universities, whose primary mission for centuries had been the preparation of young men for the ministry, and to the reactionary philosophical and religious establishment. Yet the major reason was the lack of financial support for scientific research. It was not until the 1920s that the public funding of British universities began in earnest, with the founding of the University Grants Committee.

In contrast, German universities had a strong research tradition, which could be traced back to their early exploitation of the invention of printing. They were reorganized during the Napoleonic period by Wilhelm von Humboldt (1767–1835), then head of the newly created section of culture and education of the Prussian Ministry of the Interior. Humboldt believed that university professors should excel in both teaching and research, which he held to be mutually enhancing, and set about creating the institutional conditions necessary to support this ideal. Nineteenth-century German universities were committed to the principles of *Lehrfreiheit* and *Lernfreiheit*: the freedom of professors to teach what they like and the freedom of students to study what they like (Dobson & Bruce, 1972).

The Prussian and later the unified German state provided substantial financial support for the development of German universities, which insured a well-paid professoriate and liberal grants for laboratories, books, and equipment. The epitome of the new German university was the University of Berlin, founded by von Humboldt in 1810. As a new institution, it had no ties to tradition or religious authority and spearheaded the revolution in German university education in the 19th century.

The tradition of state-supported excellence in teaching and research established at German universities was enormously influential. The system of professional institutes, chairs, and research seminars provided the model for the modern university. It encouraged the creation of specialized disciplines, including newly emerging ones such as physiology and psychology, and promoted the treatment of the doctoral degree as the qualification for university teaching. The rapid growth of German universities in the 19th century, whose expansion was seen as a prerequisite of the modern industrial state, largely accounts for the many distinguished achievements of German science in the 19th century, including experimental physiology.

Johannes Müller: Experimental Physiology

The major figure in the development of 19th-century German experimental physiology was Johannes Müller (1801–1858). He received his doctorate from the University of Bonn in 1822 and was appointed chair in physiology at the University of Berlin in 1833. A highly productive scholar, Müller did much to establish experimental physiology as an autonomous scientific discipline in Germany and Europe. His two-volume *Handbook of Human Physiology* (1833–1840) became the internationally recognized sourcebook of contemporary research in physiology and neuroanatomy for generations of researchers, replacing von Haller's earlier compendium.

Müller made many important contributions to experimental physiology. He replicated Hall's studies of the reflex arc and Magendie's experimental discrimination of sensory and motor nerves in the spinal cord. He developed an integrated hierarchical theory of the functions of the nervous system and developed an early account of "trial-and-error" learning based upon spontaneous nervous activity, which was probably the source of Bain's account of voluntary behavior.

Following a speculation by Bell, Müller demonstrated that there are five types of sensory nerves, each with its own "specific energy," which give rise to distinctive sensations of color, smell, taste, sound, and touch. Müller believed that he was investigating the physiological basis of the Kantian categories and claimed that the distinctive properties of our sensations of color, smell, taste, sound, and touch are determined by our nervous system, although he was never sure whether specific nerves or the areas of the brain to which they project are responsible for them.

Perhaps Müller's greatest achievement was as a teacher. He inspired many distinguished students, such as Ernst W. von Brücke (1819–1892), Emil du Bois-Reymond (1818–1896), Carl F. W. Ludwig (1816–1895), Hermann von Helmholtz (1821–1894), and Theodor A. H. Schwann (1810–1882), who made significant contributions to experimental physiology and physiological psychology. A workaholic prone to depression, Müller is believed to have taken his own life when he grew fearful of his declining powers (Young, 1990).

Vitalism and the Berlin Physical Society Müller was in an important sense the inheritor of the mechanistic approach to animal and human psychology and behavior initiated by Descartes. And like Descartes, Müller was a champion of **vitalism**. Descartes had taken the revolutionary step of separating the principles of life and mind that had been equated by ancient and medieval theorists. He had maintained that vital processes such as respiration and digestion are a mechanical product of organized matter, rather than a product of the action of the rational soul.

Yet by the 19th century vitalism had developed into the view that physiological processes are the product of an emergent vital force *distinct* from the physical and chemical forces of attraction and repulsion. This was the view held by Xavier Bichat (1771–1802), who claimed that vital processes are not reducible to the laws of physics and chemistry, and the chemist Justus von Liebig (1803–1873), who treated vital force as "a peculiar property, which is possessed by certain material bodies and becomes sensible when their elementary particles are combined in a certain arrangement or form" (quoted in Lowry, 1982, pp. 71–72).

In the early 19th century, it was common to appeal to vital force to explain how physical-chemical forces binding the constituents of food are overcome in the process of digestion:

> The vital force causes a decomposition of the constituents of food, and destroys the force of attraction which is continually exerted between their molecules; it alters the direction of the chemical forces in such wise, that the elements of the constituents of food arrange themselves in another form. . . . It causes the new compounds to assume forms altogether different from those which are a result of the attraction of particles when acting freely, that is, without resistance. . . . The phenomenon of growth, or increase in the mass, presupposes that the acting vital force is more powerful than the resistance which the chemical force opposes to the decomposition or transformation of the elements of the food.
>
> —(quoted in Lowry, 1982, p. 71)

As Johann F. Blumenbach (1752–1840) stressed, vital force was postulated, like gravitational force, on the basis of its observed effects. So long as there remained physiological processes that could not be reductively explained in terms of the known forces of physics and chemistry, it was reasonable to postulate such a force.

This was the form of vitalism that Müller championed. However, his students would have none of it. In 1842, Brücke and du Bois-Reymond reported a solemn oath, sealed in blood, which they had taken with Ludwig and Helmholtz, to the effect that: "no other forces than the common physical-chemical ones are active in the organism" (du Bois-Reymond, 1842/1997, p. 19). They founded the **Berlin Physical Society** in 1845, dedicated to the reductive explanation of physiological processes. They all went on to hold major chairs at German universities.

Their commitment to reductive explanation was empirically validated by Ludwig's account of the formation of urine, the first detailed explanation of a physiological process in terms of a well-understood physical-chemical process (Boakes, 1984). The daunting complexity of most other physiological processes precluded the systematic reduction of the physiological to the physical-chemical in the 19th century (Cranefield, 1957), but the commitment of Müller's students to reductive explanation inspired them to make substantive contributions to the study of nervous transmission and reflexive behavior.

Yet it would be wrong to suppose that vitalism impeded the development of experimental physiology. Müller and fellow vitalists such as Claude Bernard (1813–1878) and Louis Pasteur (1822–1895) were gifted experimentalists who made substantive contributions to 19th-century physiology, just as dualists such as Flourens and Fritsch and Hitzig made significant contributions to the neural localization of psychological capacities. However, Müller's commitment to vitalism may partly explain the reluctance of many theorists to embrace his account of voluntary behavior, based upon the spontaneous activity of the nervous system.

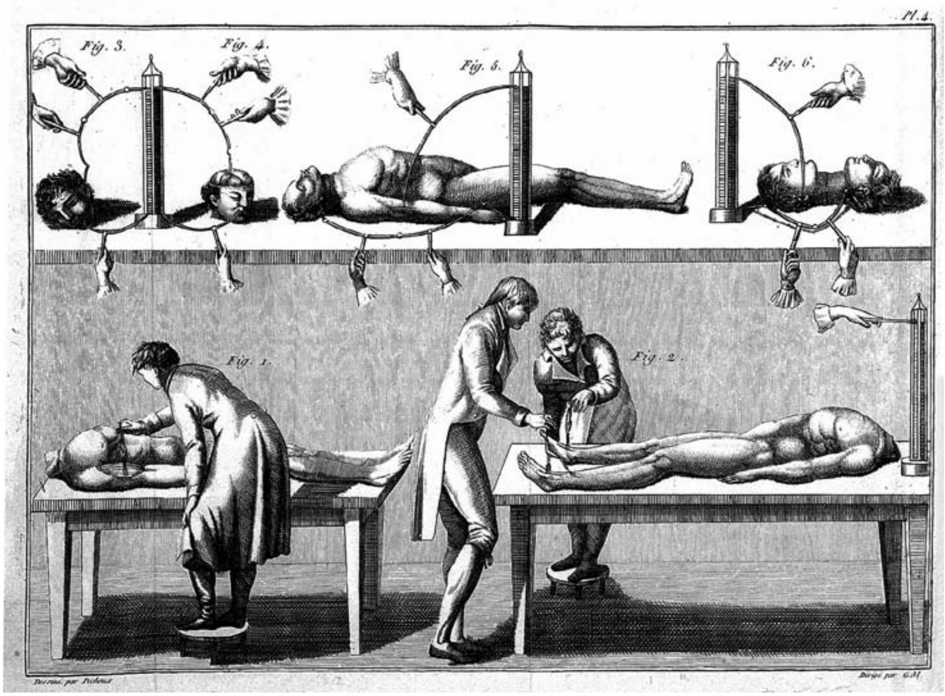
Emil du Bois-Reymond: Electrophysiology

Electrical phenomena were the subject of great interest in the 18th and 19th century. Benjamin Franklin's (1706–1790) dramatic experiments with static electricity and his explanation of lightning were enthusiastically received in Europe and America. Popular demonstrations of the ability of the human body to serve as an electrical conductor led many to speculate about the role of electricity in physiology and psychology. One of the most popular scientific texts in mid-19th-century America was the Reverend John Bovee Dods's 1850 book *Electrical Psychology*, and electrotherapy was a common form of medical treatment in the late 19th century (Reed, 1997). William James recommended it for his sister Alice and himself.

In the early 18th century, Hales had speculated that electricity might be the elusive force behind nervous action, the "vis nervosa" about which Whytt had admitted ignorance. In the late 18th century Luigi Galvani (1737–1798) claimed to have demonstrated the electrical nature of nervous activity by producing contractions in the leg muscles of frogs, which he connected to different metallic elements. His nephew Giovanni Aldini engaged in more dramatic demonstrations by electrically inducing spasmodic muscular responses in the severed heads of criminals (Boakes, 1984).

Alessandro Volta (1745–1827) disputed Galvani's results. He claimed that Galvani had only identified a form of "metallic electricity" based upon the potential (or voltage) difference between two metals. The ensuing controversy was remarkably productive for the development of electrical theory and electrophysiology. During the following decades, more sophisticated technical devices for the electrical stimulation of living tissue were created, and finely calibrated instruments such as galvanometers enabled physiologists to measure very small amounts of electricity (Boakes, 1984).

Galvani and Volta were both vitalists, who believed that they were measuring the relation between electrical energy and vital force (Reed, 1997). Müller rejected the notion that the "vis nervosa" is electrical in nature, but his student du Bois-Reymond provided experimental evidence for the electrical basis of neural transmission. Du Bois-Reymond, who took over Müller's chair in physiology at



Aldini: electrical stimulation of the brains of criminals.

the University of Berlin, demonstrated that the nervous system conducts rather than generates electricity (as Galvani and Volta had maintained) and that every nervous tissue (and not merely muscular tissue) contains an electromotive force or "resting potential" (Boakes, 1984). His pioneering experimental studies were published in *Animal Electricity* (1848–1849).

Just how the nervous system conducted electricity remained a mystery, since it seemed a poor candidate for a conductive device. It was well known that a metal wire could conduct electricity as long as it was insulated, but the nervous system seemed to lack insulation, and its wet tissues appeared to guarantee the immediate dissipation of any electrical charge. The modern understanding of electrical transmission along individual cells—or neurons—only came about with the development of the cell theory, originally advanced by Theodor Schwann, another of Müller's students, and established by the Spanish physiologist Ramon y Cajal (1852–1934) toward the end of the 19th century (Boakes, 1984).

The notion that neural transmission is a form of electrical conduction had one theoretical virtue. Transmission of electrical current in an insulated conductor is very fast (close to the speed of light), which would explain the speed of executed human decisions. Our decision to wave to a friend, for example, is almost instantaneously followed by our arm rising, despite transmitted signals having to travel the length of nervous tissue linking the brain and arm muscles (Boakes, 1984). Müller, who rejected the electrical theory of nervous conduction, maintained that it was too fast to be measured, a claim that was quickly falsified by the experimental work of yet another of his students, Hermann von Helmholtz.

Hermann von Helmholtz: Physiological Psychology

Hermann von Helmholtz (1821–1894) made major contributions to physics as well as physiology and physiological psychology. An army surgeon who was honorably relieved of his duties so he could devote himself full-time to his scientific research, Helmholtz taught at the universities of Berlin, Königsberg, Bonn, and Heidelberg. In a famous paper produced in 1847, he advanced the principle of the conservation of energy, according to which the total quantity of energy remains constant throughout any qualitative change. The principle was held to be universal in scope, applying to physical, chemical, physiological, and—by implication—psychological systems. According to it, the physical world, including living beings and their psychologies, constitutes a closed system. To postulate a psychic or vital force that is categorically distinct from physical energy would be to violate the principle of "closed physical causality" and threaten the possibility of a law-governed science of physiology (O'Donnell, 1985).

This did not pose as much a threat to psychic or vital force as might be supposed, given the recognition of the exchangeability of physical forces, such as the conversion of heat to mechanical energy. Psychic force came to be conceived as a special form of physical or electrical force. For Helmholtz and his colleagues, this suggested that conscious experience could be identified with the transformation of energy traveling through the nervous system. The nervous system came to be represented primarily as a conductor of electrical energy, received via stimulation of sensory receptors and discharged through motor behavior, analogous to the recently developed telegraph (Lenoir, 1994).

Helmholtz was the first to measure the speed of neural conduction. He estimated it at around 25–45 meters per second in frogs and around 30–35 meters per second in humans. He demonstrated that the speed of neural conduction varies with distance from the central nervous system and that it is too slow to be purely electrical in nature. It was later determined to be electrochemical in nature, largely through the work of Thomas R. Elliott (1877–1961), Henry Dale (1875–1968), and Otto Loewi (1873–1961).

Perception as Unconscious Inference Helmholtz also focused his attention on the problem that had concerned Berkeley 150 years earlier. Helmholtz, like Müller and many of his contemporaries, postulated a system of **punctiform sensations** (analogous to the atomistic sense impressions of the British empiricists) as the basis of complex perception. Individual receptors (in the retina, for example, in the case of vision) were held to carry sensory excitation along discrete neural pathways to individual projection areas in the brain. Yet it was recognized that we do not have punctiform sensations of distance, shape, size, causality, motion, and the like, and that our perception of these properties is more than the mere aggregation or association of punctiform sensations. Helmholtz claimed that our perception of these properties is based upon an unconscious *cognitive* inference.

Like Berkeley, Helmholtz insisted that distance perception is a product of empirical learning, but recognized that learning alone cannot explain how punctiform sensations get transformed into unified perceptions of the distance of physical bodies. According to Helmholtz, we have innate ideas of distance, shape, size, causality, motion, and the like that we correlate with sensory experience to yield cognitive judgments about physical bodies and their properties based upon inference (Turner, 1977, 1982):

> If a connection is to be formed between the idea of a body of certain figure and certain position, and our sensations of sense, then we first have to have the idea of such bodies. Just as with the eye, so it is also with the other senses; we never perceive the objects of the external world directly, on the contrary, we only perceive the effects of these objects on our nervous apparatuses, and it always has been like that from the first moment of our life. Now, in which way have we passed over for the first time from the world of sensations of our nerves to the world of reality? Obviously only through an inference.
>
> —(1855, p. 40, trans. Pastore, 1974)

Although our "perception" of distance, shape, size, causality, motion, and the like is really a cognitive judgment based upon inference from repeated sensory experience, it appears as a form of direct perception because it is unconscious and instantaneous. Helmholtz's theory anticipated the general form of many theories in late-20th-century cognitive psychology.

Helmholtz also developed a trichromatic theory of color vision (based upon the three primary colors), now known as the Young-Helmholtz theory of color vision, since it was developed independently by Thomas Young (1773–1829) in 1802. He also made important contributions to acoustics. His *Treatise on Physiological Optics* was published in 1856–1866 and his *On the Sensation of Tone* in 1862.

Ivan Sechenov: Inhibition

Theories of reflex behavior from Descartes to Müller had presupposed that reflexive behavior is grounded in the *excitation* of the nervous system, with energy from stimulated sensory receptors being conducted through the nervous system to generate

Ivan Sechenov and frogs.

motor responses. In 1845 the German physiologist Edouard Weber (1806–1871) of the University of Leipzig made a major discovery that eventually transformed theories of neurophysiological function. Weber demonstrated that stimulation of the vagus nerve (which runs from the brain to various internal organs) leads to a *reduction* in heart rate: this was the first experimental demonstration of increased activity in one part of the nervous system leading to *decreased* activity in another part. Weber demonstrated that the nervous system functions to *inhibit* as well as stimulate behavior (Boakes, 1984).

The significance of **inhibition** was not immediately appreciated. Edouard F. W. Pflüger (1829–1910), yet another of Müller's students, demonstrated that neural stimulation can inhibit activity in the intestine of a frog. Pflüger did not attach any special significance to neural inhibition. However, his experimental report was carefully studied by Ivan Mikhailovich Sechenov (1829–1905), a Russian student newly arrived at the University of Berlin.

Ivan Sechenov, the founder of Russian reflexology, studied with Müller, du Bois-Reymond, Ludwig, and Helmholtz. He later become professor of physiology at the Military-Medical Academy of the University of St. Petersburg and published *Reflexes of the Brain* in 1863. On the basis of experiments conducted on frogs, Sechenov demonstrated that stimulation of certain regions of the brain (for example, regions of the thalamus) depresses normal reflex activity, such as a frog's automatic withdrawal of its leg when placed in diluted acid. Sechenov was aware that many automatic reactions, such as sneezing and coughing, can be voluntarily suppressed, and he theorized that voluntary behavior is reflexive behavior that has come under the control of inhibitory stimuli. According to Sechenov, there are neural mechanisms that serve both to inhibit and to enhance reflexive behavior and that become associated with behavior through established habits. What is commonly conceived of as a strong will is simply the product of successfully learned inhibition or enhancement of reflexive behavior, such as the ability to refrain from alcohol or to increase one's speed in a competitive race.

Sechenov's account of voluntary behavior was thoroughly mechanistic: He treated all behavior as a function of innate and learned reflexes and learned inhibition and enhancement. He rejected Müller's account of voluntary behavior as a product of the spontaneous activity of the brain, because he associated the notion of spontaneous activity with vitalism. Sechenov claimed that all behavior is a causal product of sensory stimulation, since otherwise energy sufficient to produce behavior would have to come from some source outside the nervous system, such as an immaterial soul or vital force. He rejected explanations of behavior in terms of internal mental states such as thought and desire, because he believed that these are merely links in a causal chain running from sensory stimulation to (reflexive) behavior:

> Thought is generally believed to be the cause of behavior . . . but this is the greatest of falsehoods; the initial cause of all behavior always lies, not in thought, but in external sensory stimulation, without which no thought is possible.
>
> —(1863/1965, p. 322)

Yet Sechenov was no epiphenomenalist. He did not deny that behavior is a causal product of thought and desire. Rather, he maintained that thought and desire are merely the **proximate** or immediate causes of behavior, which are themselves fully determined by external sensory stimulation. He suggested that contemplative thought is a reflex in which the final behavioral outcome is suppressed through learned inhibition (Smith, 1992):

> Now, a psychical act . . . cannot appear in consciousness without an external sensory stimulation. Consequently, our thoughts are also subject to this law; therefore, in a thought, we have the beginning of a reflex, and its continuation; only the end of a reflex (i.e. the movement) is apparently absent.
>
> *A thought is the first two thirds of a psychical reflex.*
>
> —(Sechenov, 1863/1965, pp. 320–321)

Sechenov's extreme position on this matter may have been a consequence of his independent commitment to an extreme environmentalism. He was a political radical who hoped that psychology would enable humans to realize their true potential and surmount the repressive constraints of traditional societies, such as the Tzarist Russia to which he returned. He was not driven to this position by his rejection of vitalism, because there is no intrinsic connection between Müller's account of voluntary behavior as the product of spontaneous neural activity and vitalist theories of physiological function. Both Bain in the 19th century and B. F. Skinner in the 20th century developed nonreflexive accounts of learned behavior that did not presuppose any commitment to vitalism. They developed accounts of learned behavior as originally spontaneous (Bain) or random (Skinner) behavior that is transformed into directed behavior through association with pleasure or reinforcement, *independently* of sensory stimulation.

Sechenov also may have been influenced by the popular conception of the central nervous system as a conductive device, analogous to the telegraph, through which electrical energy generated by the stimulation of sensory receptors is transformed into behavioral responses. This conception of the nervous system certainly shaped the German tradition of research on **psychophysics**, the study of the relation between the objective intensity of physical stimuli and subjective sensational experience.

Gustav Fechner: Psychophysics

The German physicist Gustav Theodor Fechner (1801–1887) provided the initial link between the experimental physiology of the 19th century and the experimental psychology of the late 19th and early 20th centuries. He earned a medical degree at the University of Leipzig in 1822, but his main interests lay in physics and mathematics. He became professor of physics at the University of Leipzig, where he did significant research on the measurement of electric current. Fechner used direct sunlight as a stimulus for his studies of visual after-images, with himself as experimental subject. He injured his eyes so badly that he was forced to resign his position at the university, although he returned a few years later.

Fechner believed that mental and physical states and processes are qualitatively different but quantitatively identical. Although they appear different, they are ultimately one and the same. After a number of years of depression and physical illness following his optical injury, Fechner made a dramatic recovery when he suddenly realized how he could establish the identity of the mental and the physical. He could make the "relative increase of bodily energy the measure of the increase of the corresponding mental intensity" (1860/1966, p. 3).

Assuming their identity, and Helmholtz's conservation of energy principle, Fechner reasoned that mental and physical processes must be functionally related. He also assumed they must be governed by laws of proportional variation rather than simple covariation, given the fact of resistance in any electrical system, including the nervous system. In a series of experiments, he set out to determine the mathematical laws governing this functional relationship. He systematically varied the intensity of (auditory, visual, tactual, and thermal) stimuli, and measured the intensity of sensational responses by means of **just noticeable differences** between the perceived intensity of sensations. Fechner concluded that the perceived intensity of a sensation is a logarithmic function of the physical intensity of a stimulus. This relationship is expressed in the formula now known as **Fechner's law**: $S = \mathrm{k} \log R$ (where R represents the physical intensity of a stimulus, S the perceived intensity of a sensation, and k is a constant). Fechner's law was

Psychophysics experiment: weight estimation.

a mathematical transformation of the ratio between the intensity of a physical stimulus and the perceived intensity of sensation established by his colleague, Ernst Weber (1795–1878), sometimes known as Weber's law.

Fechner's studies of the relationship between the intensity of physical stimuli and the perceived intensity of sensational responses were published in *Elements of Psychophysics* in 1860. William James dismissed his "dreadful" contribution as amounting to "nothing." However, many were convinced that Fechner had refuted Kant's claim that psychology could not attain the status of a genuine science, since he had established quantified psychophysical laws based upon experiments in which manipulated differences in the physical intensity of stimuli were correlated with subjects' introspective reports of sensational differences.

Because of his pioneering psychophysical studies, Fechner is often represented as having established the physical basis of mentality by demonstrating the functional dependence of the mental on the physical. There is some irony in this, since Fechner himself believed he had demonstrated the opposite: He believed that he had demonstrated the mentality of the physical and the existence of a "world-soul" (Reed, 1997). Under the pen name of Dr. Mises, he railed against the materialism of his age.

Boring (1957) called Fechner the founder of scientific psychology, but this is an exaggeration. Although Fechner did extend the experimental methods of physiology to psychology by developing psychophysics, his own experimental work was restricted to psychophysics, and he played no significant role in the institutional development of scientific psychology. However, Fechner's development of psychophysics was the first step in the evolution of a form of physiological psychology distinct from experimental physiology. The experimental studies of late-19th-century physiology were generally restricted to the electrophysiology of the nervous system, the physiology of the sensory organs, and the integration of motor reflexes. Although it was relatively easy to map the motor cortex of animals, it was much harder to map their sensory cortex, since differences in the sensory responses of animals are difficult to determine empirically. In order to develop the sensory-motor theory of the nervous system, the experimental method had to be extended to the introspective study of the nature of sensation, the historical domain of human psychology.

PHYSIOLOGICAL PSYCHOLOGY AND OBJECTIVE PSYCHOLOGY

Although German scientific psychology grew out of the achievements of 19th-century experimental physiology, it was not restricted to the types of reductive physiological explanation of psychological processes favored by the members of the Berlin Physical Society or to the psychophysical studies pioneered by Fechner. Wilhem Wundt, who founded scientific psychology in Germany in the late 19th century, insisted on the autonomy of psychological explanation with respect to physiological explanation. He called his form of experimental psychology **physiological psychology** because it was based upon the experimental methods of physiology, not because it was based upon the explanatory concepts of physiology.

In contrast, Sechenov advocated an **objective psychology** that was based not only upon the experimental methods of physiology but also upon its explanatory concepts. Sechenov's *Reflexes of the Brain* was originally titled *An Attempt to Bring Physiological Bases Into Mental Processes*. In his later article *Who Must Investigate the Problems of Psychology and How* (1871/1973), he maintained that progress in psychology could be achieved only by developing reflexive physiological theories of human and animal behavior.

Sechenov was committed to the strong continuity of human and animal psychology. He consequently maintained that the study of human psychology is best approached through the study of animal psychology, in which the basic reflexive components of human psychology are revealed in their elemental form:

> It is clear then that the psychical phenomena of animals, and not those of man, should be used as the primary material for studying psychical phenomena.
>
> —(1871/1973, p. 339)

This principle, which became the foundation of 20th-century behaviorist psychology, received powerful support from 19th-century developments in evolutionary theory.

DISCUSSION QUESTIONS

1. Try to make a case for Comte's claim that all of psychology is exhausted by sociology and biology. Do you find this convincing?
2. Mill thought that psychology was bound to remain an inexact science, and of limited predictive utility, because of the difficulty of anticipating the complex conditions of human behavior. Is psychology really different from other sciences in this respect?
3. Suppose (with Dr. Molyneux in the 18th century) that a man born blind, who learned to navigate his surroundings, gained his sight through an operation in later years. What would Berkeley predict with respect to his ability to perceive distance? What would Bailey predict? What would Mill predict?
4. According to Bain and Müller, voluntary behavior is a form of learned behavior that organisms originally generate spontaneously, that is, independently of sensory stimulation. Why do you think that early psychologists dismissed this account as unscientific? Is it unscientific?
5. Gall maintained that the 27 fundamental psychological capacities he had identified and located in various regions of the brain were distinct and independent. Would it have created any special problems for neural localization if he had conceived of these capacities relationally rather than atomistically?
6. Consider the relations between reflexive, purposive, conscious, and voluntary behavior. Can reflexive behavior be unconscious but purposive? Can behavior be reflexive and conscious? Can voluntary behavior be reflexive? Does voluntary behavior have to be conscious?

GLOSSARY

ablation Experimental method in physiology involving the systematic removal of neural tissue from live animals, in order to determine the function of the extirpated part of the nervous system.

ampliative inference Inference that goes beyond the information given. According to John Stuart Mill, distance perception involves an inference that goes beyond the information provided by sensation.

Bell-Magendie law The anatomical separation of sensory and motor nerves in the spinal cord, first identified by Charles Bell and experimentally confirmed by François Magendie.

Berlin Physical Society Society founded in 1845 by students of Müller opposed to vitalism, who maintained that all physiological processes can be reductively explained in terms of known physical-chemical processes.

Broca's area Superior region of the left frontal lobe of the cerebral cortex, which Broca identified as the location of the "faculty of articulate language."

conscious automata Term used by Thomas Huxley to describe humans and animals, in accord with his view that mentality and consciousness are merely epiphenomenal by-products of the reflexive mechanisms of the nervous system and play no role in the generation of animal or human behavior.

cranioscopy Phrenological identification of psychological faculties via the measurement of the contours of the skull.

epiphenomenalism Theory that mentality and consciousness are by-products of the reflexive neurophysiological states that mediate sensory-motor connections and are not causes of behavior.

ethology According to John Stuart Mill, the science of character. Mill believed that the social capacities and propensities that constitute human character could be derived from the fundamental laws of associationist psychology.

Fechner's law $S = k \log R$ (where R represents the physical intensity of a stimulus, S the perceived intensity of a sensation, and k is a constant).

ideomotor behavior Term introduced by William Carpenter to describe automatic but apparently purposive behavior that is mediated by ideas, based upon the prior association of ideas and behavior.

inhibition The ability of the nervous system to inhibit as well as stimulate activity; the ability of neural stimuli to inhibit normal reflex activity.

irritability The ability of nerves to transmit excitation resulting in muscle contraction or sensation. In the early 19th century, it was commonly believed that the cerebral cortex is not irritable or "excitable."

just noticeable difference In psychophysics, the subjective unit of measurement of the perceived difference in the intensity of a sensation.

law of three stages Comte's theory that societies pass through three stages of cognitive development—the theological, the metaphysical, and the positive—which represent fundamentally different attitudes to the explanation of natural events.

mental chemistry Term employed by John Stuart Mill to describe those association processes that are more closely analogous to chemical bonding than of mechanical combination.

metaphysical stage Second in Comte's law of three stages, in which natural events are explained in terms of depersonalized forces.

neurophysiological dualism Early-19th-century view that the cognitive functions of the cerebral cortex are categorically distinct from the sensory-motor functions of the lower brain and spinal cord.

objective psychology Sechenov's form of psychology based upon the explanatory concepts of physiology.

phrenology Theory developed by Franz Joseph Gall and Johann Casper Spurzheim, according to which the degree of development of psychological faculties is a function of the size of the area of the brain in which they are localized, which is reflected by protrusions and indentations of the skull.

physiological psychology Wundt's form of experimental psychology based upon the experimental methods of physiology, but not committed to the reductive physiological explanation of psychological processes.

points of consciousness Term introduced by James Mill to describe the discrete sensational elements of complex ideas and associations.

positive stage Last in Comte's law of three stages, in which natural events are explained in terms of the description of observable correlation.

positivism Comte's view that the highest form of human knowledge is knowledge of the correlation of observables.

proximate cause Immediate or precipitating cause.

psychophysics Study of the functional relationship between the physical intensity of stimuli and the perceived intensity of sensation.

punctiform sensations Discrete sensations (and associated neural excitations) that many 19th-century physiologists postulated as the atomistic basis of complex perception.

rational unconscious Unconscious inference governed by norms of rationality and logical inference, first postulated by John Stuart Mill in his explanation of complex perception.

reflex arc Term introduced by the English physiologist Marshall Hall to describe an elementary reflex system comprising a sensory nerve, interconnecting nervous tissue in the spinal cord, and a motor nerve.

sensory-motor theory Theory of the nervous system as a reflexive sensory-motor system whose every component can be characterized as having a sensory or motor function.

Spencer-Bain principle Theory that behavior followed by success, satisfaction, or pleasure tends to be repeated.

theological stage First in Comte's law of three stages, in which natural events are explained in terms of anthropomorphized forces.

unconscious cerebration Term introduced by William Carpenter to describe unconscious thought processes and employed by John Stuart Mill to describe unconscious inference in perception.

utilitarianism Theory that the right course of action in any situation is the one that maximizes human happiness and minimizes human misery.

vitalism Originally the view that vital processes such as respiration and digestion are a mechanical product of organized matter, rather than a product of the action of the rational soul. By the late 18th and 19th centuries it had developed into the view that physiological processes are the product of an emergent vital force *distinct* from physical and chemical forces of attraction and repulsion.

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CHAPTER 7

Theories of Evolution

THEORIES OF EVOLUTION DOMINATED INTELLECTUAL DEBATE IN Europe and America in the latter half of the 19th century, especially after the publication of Charles Darwin's *On the Origin of Species by Means of Natural Selection* (1859). Although religious authorities resisted, natural scientists and the educated public generally embraced theories of evolution: Such theories often represented human progress—or, at least, white, male, and Western human progress—as a triumph of the "survival of the fittest."

Early Greek thinkers such as Empedocles had advanced theories of the evolution of biological species. However, most scholars during the medieval period had accepted the Aristotelian account of an immutable and hierarchical natural order, or "scala naturae." Such an account not only sustained the popular conception of a purposive natural order created by a benevolent God, but also conveniently supported the notion of an immutable social hierarchy governed by kings, bishops, and the aristocracy. This account was generally accepted until the late 18th century and—by a good many theorists—beyond. The Enlightenment theories of social development and change advanced by Helvetius and Adam Ferguson (1723–1816) presupposed a fixed human nature, and Gall's phrenology presupposed a more or less fixed hierarchy of neurological function (Young, 1990).

Nineteenth-century evolutionary theorists abandoned the notion of an immutable "great chain of being" (Lovejoy, 1936) and developed explanations of the accepted fact of species change. They generally represented evolution as a process of progressive development toward a hierarchical natural order. Jean-Baptiste Lamarck (1744–1829) and Herbert Spencer (1820–1903) advanced theories that replaced the extrinsic teleology of a divinely created natural order with the intrinsic teleology of progressive development toward a natural order. Charles Darwin (1809–1882) was the exception. His rigorously materialist theory of evolution by natural selection treated evolution as a purely mechanistic process with no extrinsic or intrinsic purpose.